LAPORAN PRAKTIKUM
PEMROGRAMAN BERORIENTASI OBJEK



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PROGRAM STUDI D3 TEKNIK INFORMATIKA
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Vocabulary:

HashSet	A set similar to an ArrayList without any specific ordering. An ordered Collection that may contain duplicates. An interface used to define a group of objects. This includes lists and sets. A list that is very similar to an array. A Collection of elements that does not contain any duplicates.
List	A set similar to an ArrayList without any specific ordering. An ordered Collection that may contain duplicates. An interface used to define a group of objects. This includes lists and sets. A list that is very similar to an array. A Collection of elements that does not contain any duplicates.
Collection	A set similar to an ArrayList without any specific ordering. An ordered Collection that may contain duplicates. An interface used to define a group of objects. This includes lists and sets. A list that is very similar to an array. A Collection of elements that does not contain any duplicates.
ArrayList	A set similar to an ArrayList without any specific ordering. An ordered Collection that may contain duplicates. An interface used to define a group of objects. This includes lists and sets. A list that is very similar to an array. A Collection of elements that does not contain any duplicates.
Set	A set similar to an ArrayList without any specific ordering. An ordered Collection that may contain duplicates. An interface used to define a group of objects. This includes lists and sets. A list that is very similar to an array. A Collection of elements that does not contain any duplicates.

Try It / Solve It:

1. What is the difference between a set and a list?
   Jawaban:

- **List**: berurutan, boleh ada elemen duplikat, dan bisa tumbuh atau menyusut saat elemen ditambah atau dihapus.
- **Set**: tidak boleh ada duplikat dan tidak punya urutan maupun indeks, sehingga tidak bisa di-sort.

2. You decide you want to roll 2 dice and see what the frequency is of each possible number combination. Would you use a Set collection to do this? State your reason(s).
   Jawaban:
   Tidak, set tidak berisi duplikat, sedangkan hasil lemparan dadu pasti sering sama (misalnya jumlah 7 muncul berkali-kali). Kalau memakai Set, kemunculan yang berulang akan hilang sehingga frekuensinya tidak bisa dihitung. Pakai List (ArrayList), karena list boleh berisi duplikat.

3. Using a collection create a variable that will store a list of countries (Strings). Your collection should not store duplicates, and order is not important. Test your code by

adding 6 countries, one of which is a duplicate.
Jawaban:
Pakai HashSet, HashSet cocok dipakai saat urutan tidak penting dan set tanpa duplikat. Metode yang dipakai:

- add(e): menambah elemen kalau belum ada. Tambahan "Japan" yang kedua mengembalikan false.
- size(): jumlah elemen. Hasilnya 5, bukan 6.
- contains(e): mencari elemen di set.

(Jawaban code ada pada file Countires.java)

4. Would the following Collection.sort() statements both work? Explain your answer.

```
HashSet countriesSet = new HashSet();

Collections.sort (countriesSet);

ArrayList countriesList = new ArrayList() ;

Collections.sort(countriesList);
```

Jawaban:

- Collections.sort(countriesSet) tidak bisa. Slide 18 menyebut sort menerima List, sedangkan slide 21 menyebut HashSet tidak punya urutan atau indeks sehingga tidak bisa di-sort.
- Collections.sort(countriesList) bisa. Slide 18 menyebut “An ArrayList is a List”, jadi cocok dengan parameter sort.

5. Below is a user implementation of a Stack using arrays.
    - push adds an item to the Stack
    - pop removes an item from the stack
    - isEmpty return a Boolean value of true if the Stack is empty

   Convert this to a generic implementation using an ArrayList.

(Jawaban code ada pada file ArrayStack.java)